

**POE PART 1**

**MODULE NAME: PROGRAMMING 2B**

**MODULE CODE: PROG6212**

**STUDENT NAME: ONALERONA MOGODI**

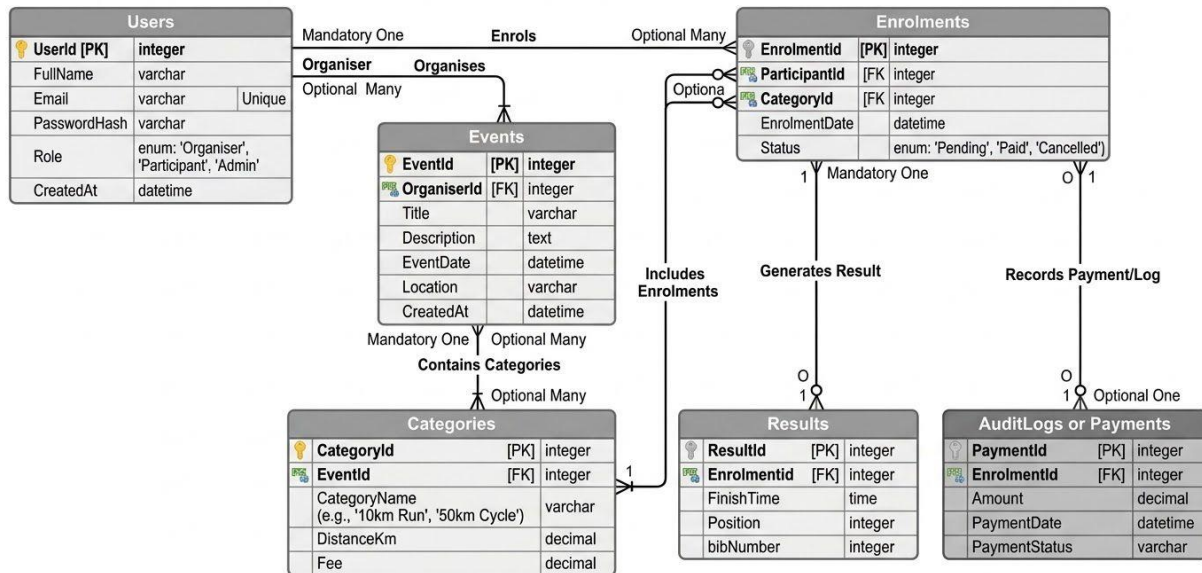
**STUDENT NUMBER: ST10467369**

**GROUP: 1**

**DUE DATE: 4 SEPTEMBER 2026**

## SECTION A

### RaceDay System: Entity-Relationship Diagram (ERD)



- Users to Events (1 : N): One Organiser creates many Events; each Event belongs to one Organiser (OrganiserId).
- Events to Categories (1 : N): One Event offers many Categories; each Category links to one Event (EventId).
- Categories to Enrolments (1 : N): One Category receives many Enrolments; each Enrolment targets one Category (CategoryId).
- Users to Enrolments (1 : N): One Participant enters many Enrolments; each Enrolment belongs to one Participant (ParticipantId).
- Enrolments to Results (1 : 0..1): An Enrolment generates zero or one Result (EnrolmentId).
- Enrolments to Payments (1 : 0..1): An Enrolment generates zero or one Payment record (EnrolmentId).

## SECTION B

| HTTP Method    | Route              | Description                                          | Role Required  | Request Body                                | Expected Response                                       |
|----------------|--------------------|------------------------------------------------------|----------------|---------------------------------------------|---------------------------------------------------------|
| Authentication |                    |                                                      |                |                                             |                                                         |
| POST           | /api/auth/register | Registers a new user account in the system.          | None (Public)  | { "fullName", "email", "password", "role" } | 201 Created - { "userId", "fullName", "email", "role" } |
| POST           | /api/auth/login    | Authenticates credentials and returns a JWT token.   | None (Public)  | { "email", "password" }                     | 200 OK - { "token", "expiresIn" }                       |
| POST           | /api/auth/logout   | Invalidates the active user session token.           | Any (LoggedIn) | None                                        | 200 OK - { "message": "Logged out successfully" }       |
| Organiser      |                    |                                                      |                |                                             |                                                         |
| GET            | /api/organisers    | Retrieves a list of all registered event organisers. | None (Public)  | None                                        | 200 OK - Array of organiser profiles                    |

|       |                         |                                                               |               |                                                   |                                                                 |
|-------|-------------------------|---------------------------------------------------------------|---------------|---------------------------------------------------|-----------------------------------------------------------------|
| GET   | /api/organisers/{id}    | Retrieves details and active events for a specific organiser. | None (Public) | None                                              | 200 OK - { "organiserId", "organizationName", "events": [...] } |
| PUT   | /api/organisers/profile | Updates the logged-in organiser's organization details.       | Organiser     | { "organizationName", "contactPhone", "website" } | 200 OK - Updated organiser profile                              |
| Venue |                         |                                                               |               |                                                   |                                                                 |
| GET   | /api/venues             | Retrieves all available event venues.                         | None (Public) | None                                              | 200 OK - Array of venue objects                                 |
| GET   | /api/venues/{id}        | Retrieves detailed information for a specific venue.          | None (Public) | None                                              | 200 OK - Venue details including capacity and address           |
| POST  | /api/venues             | Adds a new venue location to the system.                      | Organiser     | { "name", "address", "city", "capacity" }         | 201 Created - Created venue record with assigned ID             |

|        |                  |                                                                 |               |                                                    |                                                        |
|--------|------------------|-----------------------------------------------------------------|---------------|----------------------------------------------------|--------------------------------------------------------|
| PUT    | /api/venues/{id} | Updates existing venue details.                                 | Organiser     | { "name", "address", "city", "capacity" }          | 200 OK - Updated venue object                          |
| Event  |                  |                                                                 |               |                                                    |                                                        |
| GET    | /api/events      | Retrieves all upcoming race events.                             | None (Public) | None                                               | 200 OK - Array of events with venue & category summary |
| GET    | /api/events/{id} | Fetches full event details including categories and venue info. | None (Public) | None                                               | 200 OK - Full event detail object                      |
| POST   | /api/events      | Creates a new race event linked to a venue.                     | Organiser     | { "title", "description", "eventDate", "venueId" } | 201 Created - Created event details                    |
| PUT    | /api/events/{id} | Updates event information.                                      | Organiser     | { "title", "description", "eventDate", "venueId" } | 200 OK - Updated event object                          |
| DELETE | /api/events/{id} | Cancels and removes an event.                                   | Organiser     | None                                               | 204 No Content / 200 OK                                |

|           |                                  |                                                     |               |                                                        |                                                        |
|-----------|----------------------------------|-----------------------------------------------------|---------------|--------------------------------------------------------|--------------------------------------------------------|
| Category  |                                  |                                                     |               |                                                        |                                                        |
| GET       | /api/events/{eventId}/categories | Retrieves all categories for a given event.         | None (Public) | None                                                   | 200 OK - Array of categories (e.g., 10km, 42km)        |
| POST      | /api/events/{eventId}/categories | Adds a race category/distance to an event.          | Organiser     | { "categoryName", "distanceKm", "fee", "maxCapacity" } | 201 Created - Category details                         |
| PUT       | /api/categories/{id}             | Updates details for a specific category.            | Organiser     | { "categoryName", "distanceKm", "fee" }                | 200 OK - Updated category object                       |
| DELETE    | /api/categories/{id}             | Removes a category from an event.                   | Organiser     | None                                                   | 204 No Content                                         |
| Enrolment |                                  |                                                     |               |                                                        |                                                        |
| POST      | /api/categories/{id}/enrol       | Enrols the authenticated participant in a category. | Participant   | { "paymentMethod" }                                    | 201 Created - { "enrolmentId", "status", "bibNumber" } |
| GET       | /api/enrolments/{id}             | Gets details of a specific                          | Participant   | None                                                   | 200 OK - Enrolment details                             |

|             |                                    |                                                                     |             |                                         |                                                            |
|-------------|------------------------------------|---------------------------------------------------------------------|-------------|-----------------------------------------|------------------------------------------------------------|
|             |                                    | enrolment record.                                                   |             |                                         |                                                            |
| GET         | /api/enrolments/my-races           | Retrieves all enrolments for the logged-in participant.             | Participant | None                                    | 200 OK - Array of participant's enrolments                 |
| DELETE      | /api/enrolments/{id}               | Cancels an active enrolment.                                        | Participant | None                                    | 200 OK - { "message": "Enrolment cancelled" }              |
| Participant |                                    |                                                                     |             |                                         |                                                            |
| GET         | /api/participants/profile          | Fetches profile and emergency details of the logged-in participant. | Participant | None                                    | 200 OK - { "userId", "emergency Contact", "medicalNotes" } |
| PUT         | /api/participants/profile          | Updates participant profile, emergency info, or medical notes.      | Participant | { "emergency Contact", "medicalNotes" } | 200 OK - Updated participant profile                       |
| GET         | /api/events/{eventId}/participants | Lists all participant                                               | Organiser   | None                                    | 200 OK - Array of                                          |

|  |  |                                          |  |  |                                    |
|--|--|------------------------------------------|--|--|------------------------------------|
|  |  | s enrolled<br>in a<br>specific<br>event. |  |  | enrolled<br>participant<br>details |
|--|--|------------------------------------------|--|--|------------------------------------|

**SECTION C**





SSMS interface showing a SQL query execution. The query is titled "SQLQuery1.s...udent (65)" and contains the following SQL code:

```
67
68 INSERT INTO Users (FullName, Email, PasswordHash, Role) VALUES
69 ('Sarah Jenkins', 'sarah@eventpros.com', 'hashed_pass_1', 'Organiser'),
70 ('David Smith', 'david@marathonorg.co.za', 'hashed_pass_2', 'Organiser'),
71 ('John Doe', 'john.doe@gmail.com', 'hashed_pass_3', 'Participant'),
72 ('Alice Johnson', 'alice.j@gmail.com', 'hashed_pass_4', 'Participant');
73
74 INSERT INTO Events (OrganiserId, Title, Description, EventDate, Location) VALUES
75 (1, 'Gauteng City Marathon', 'Annual road running event.', '2026-10-15 06:00:00', 'Johannesburg'),
76 (1, 'Trail Run Challenge', 'Off-road trail adventure.', '2026-11-20 07:00:00', 'Pretoria'),
77 (2, '907 Cycle Challenge', 'Premier cycling endurance event.', '2026-12-05 05:30:00', 'Midrand');
78
79 INSERT INTO Categories (EventId, CategoryName, DistanceKm, Fee) VALUES
80 (1, 'Half Marathon', 21.10, 250.00),
81 (1, '10K Trail Run', 10.00, 150.00),
82 (3, '90K Open Road', 90.00, 500.00);
83
84 INSERT INTO Enrolments (ParticipantId, CategoryId, Status) VALUES
85 (3, 1, 'Confirmed'),
86 (3, 3, 'Confirmed'),
87 (4, 2, 'Confirmed'),
88 (4, 4, 'Confirmed');
89
90 INSERT INTO Results (EnrolmentId, FinishTime, Position, BibNumber) VALUES
91 (1, '03:45:12', 12, 1001),
92 (3, '01:52:30', 5, 2005)
93
```

The status bar indicates "Query executed successfully." and "100%". The bottom status bar shows "Ready" and "23°C Sunny".

SSMS interface showing a SQL query execution. The query is titled "SQLQuery4.s...udent (92)" and contains the following SQL code:

```
1 SELECT TOP (1000) [LogId]
2     , [Action]
3     , [PerformedBy]
4     , [Timestamp]
5 FROM [RaceDayDB].[dbo].[AuditLogs]
6
```

The status bar indicates "Query executed successfully." and "100%". The bottom status bar shows "Ready" and "23°C Sunny".

SQLQuery5.s...tudent (74)

```
1 SELECT TOP (1000) [CategoryId]
2     ,[EventId]
3     ,[CategoryName]
4     ,[DistanceKm]
5     ,[Fee]
6 FROM [RaceDayDB].[dbo].[Categories]
7
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

| CategoryId | EventId | CategoryName  | DistanceKm | Fee    |
|------------|---------|---------------|------------|--------|
| 1          | 1       | Full Marathon | 42.20      | 350.00 |
| 2          | 1       | Half Marathon | 21.10      | 250.00 |
| 3          | 2       | 10K Trail Run | 10.00      | 150.00 |
| 4          | 3       | 90K Open Road | 90.00      | 500.00 |

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) LabVM2049939\Student (74) RaceDayDB 00:00:00 Row: 1, Col: 1 4 rows

SQLQuery6.s...tudent (76)

```
1 SELECT TOP (1000) [EnrolmentId]
2     ,[ParticipantId]
3     ,[CategoryId]
4     ,[EnrolmentDate]
5     ,[Status]
6 FROM [RaceDayDB].[dbo].[Enrolments]
7
```

100 % No issues found Ln: 1, Ch: 1 SPC CRLF Windows 1252

| EnrolmentId | ParticipantId | CategoryId | EnrolmentDate           | Status    |
|-------------|---------------|------------|-------------------------|-----------|
| 1           | 3             | 1          | 2026-09-03 13:20:12.100 | Confirmed |
| 2           | 3             | 3          | 2026-09-03 13:20:12.100 | Confirmed |
| 3           | 4             | 2          | 2026-09-03 13:20:12.100 | Confirmed |
| 4           | 4             | 4          | 2026-09-03 13:20:12.100 | Confirmed |

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) LabVM2049939\Student (76) RaceDayDB 00:00:00 Row: 1, Col: 1 4 rows

SQLQuery7.s...tudent (80)

```
1 SELECT TOP (1000) [EventId]
2     ,[OrganiserId]
3     ,[Title]
4     ,[Description]
5     ,[EventDate]
6     ,[Location]
7     ,[CreatedAt]
8 FROM [RaceDayDB].[dbo].[Events]
9
```

100 % No issues found Ln: 1, Ck: 1 SPC CRLF Windows 1252

| EventId | OrganiserId | Title                 | Description                      | EventDate               | Location     | CreatedAt               |
|---------|-------------|-----------------------|----------------------------------|-------------------------|--------------|-------------------------|
| 1       | 1           | Gauteng City Marathon | Annual road running event.       | 2026-10-15 06:00:00.000 | Johannesburg | 2026-09-03 13:19:43.537 |
| 2       | 1           | Trail Run Challenge   | Off-road trail adventure.        | 2026-11-20 07:00:00.000 | Pretoria     | 2026-09-03 13:19:43.537 |
| 3       | 2           | 947 Cycle Challenge   | Premier cycling endurance event. | 2026-12-05 05:30:00.000 | Midrand      | 2026-09-03 13:19:43.537 |

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) LabVM2049939\Student (80) RaceDayDB 00:00:00 Row: 1, Col: 1 3 rows

SQLQuery8.s...tudent (95)

```
1 SELECT TOP (1000) [ResultId]
2     ,[EnrolmentId]
3     ,[FinishTime]
4     ,[Position]
5     ,[BibNumber]
6 FROM [RaceDayDB].[dbo].[Results]
7
```

100 % No issues found Ln: 1, Ck: 1 SPC CRLF Windows 1252

| ResultId | EnrolmentId | FinishTime       | Position | BibNumber |
|----------|-------------|------------------|----------|-----------|
| 1        | 1           | 03:45:12.0000000 | 12       | 1001      |
| 2        | 3           | 01:52:30.0000000 | 5        | 2005      |

Query executed successfully. (local)\SQLEXPRESS (17.0 RTM) LabVM2049939\Student (95) RaceDayDB 00:00:00 Row: 1, Col: 1 2 rows

SQLQuery9.s...tudent (97) | SQLQuery8.sql...Student (95) | SQLQuery7.sql...Student (80) | SQLQuery6.sql...Student (76) | SQLQuery5.sql...Student (74) | ...

```
1 SELECT TOP (1000) [UserId]
2 , [FullName]
3 , [Email]
4 , [PasswordHash]
5 , [Role]
6 , [CreatedAt]
7 FROM [RaceDayDB].[dbo].[Users]
8
```

100 % | No issues found | Ln: 1, Ch: 1 | SPC | CRLF | Windows 1252

|   | UserId | FullName      | Email                   | PasswordHash  | Role        | CreatedAt               |
|---|--------|---------------|-------------------------|---------------|-------------|-------------------------|
| 1 | 1      | Sarah Jenkins | sarah@eventpros.com     | hashed_pass_1 | Organiser   | 2026-09-03 13:19:21.827 |
| 2 | 2      | David Smith   | david@marathonorg.co.za | hashed_pass_2 | Organiser   | 2026-09-03 13:19:21.827 |
| 3 | 3      | John Doe      | john.doe@gmail.com      | hashed_pass_3 | Participant | 2026-09-03 13:19:21.827 |
| 4 | 4      | Alice Johnson | alice.j@gmail.com       | hashed_pass_4 | Participant | 2026-09-03 13:19:21.827 |

Query executed successfully. | (local)\SQLEXPRESS (17.0 RTM) | LabVM2049939/Student (97) | RaceDayDB | 00:00:00 | Row: 1, Col: 1 | 4 rows

Ready | 23°C Sunny | Search | 1:24 PM 9/3/2026

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